In sentence S2, White House is a location. Can you come up with an algorithm that can disambiguate such entity mentions to their appropriate types based on the context?

4. You have been asked to build an application that analyses the conversations in call centres. These conversations typically happen between the agent and the customer. An automatic speech recognition system has been used to convert the voice conversations into text. The first stage of your application needs to analyse the voice transcripts, and correct any mistakes in the words that may have been done by the automatic speech recognition system. Can you explain how you will build the transcript correction system?
5. You are given a corpus and have created the word embeddings for it using the popular Word2Vec technique. Now you need to represent each sentence in your corpus by means of a vector. You need to compose the word vectors of the words present in your sentence to generate a sentence vector representation. What are the possible compositional models you can apply to generate the sentence vector?
6. Consider the two sentences:
   S1: The bank will remain closed on Saturday.
   S2: He sat on the bank and watched the sunset.
   The word 'bank' has two senses, one as a commercial institution and another as the sand bank by the side of the river. The appropriate context has to be assigned to the word 'bank'. This is the problem of word sense disambiguation. Can you come up with an algorithm for word sense disambiguation using a supervised technique (provided that you are given sufficient labelled data)? How will your system handle words whose multiple senses are not present in the training data? What are the trade-offs in using a supervised vs an unsupervised approach towards word sense disambiguation?
7. You are running a randomised clinical trial, where you are asked to determine whether a newly discovered drug for cancer is effective. You have a control group of patients who do not get the new drug and a target group of patients who get the drug. Let us assume that the clinical effectiveness of the new drug is measured in terms of the reduction in number of deaths in the control and target groups for six months following the clinical trial start. Can you apply the concept of A/B testing for this situation? If no, explain why not.
8. While search engines have traditionally been tasked with the goal of returning Web pages relevant to the search query, they are increasingly being used to answer user questions. Google uses its knowledge graph to provide fact-checked information about well-known personalities along with traditional search results. For example, if you type the question, 'What is the first book by J K Rowling?', you will get a text box containing the relevant snippet as the answer along with standard search results. How will you design a search engine that can provide direct answers to factoid questions?
9. Given a large text document, you are asked to create a summary of the same. There are two types of summarization algorithms, namely, extractive vs abstractive summarization. In extractive summarization, the summary will contain sentences extracted from the text. In abstractive summarization, the summary will contain the gist of the original text, without including the exact sentences from the text. What kind of summarization algorithm will you choose? Explain the rationale behind your choice. Now assume there is a real time conversation that is occurring. How will you create a running summary of the same?
10. Let us consider a binary classification problem, where you are asked to label emails as spam or non-spam. You are given a large enough training data set (100,000 emails), where the emails are already classified as spam or non-spam. Using this training data, you are asked to train a binary classifier that can mark an incoming email as spam or non-spam. However, there is one problem with the training data that you had been given. Among the 100,000 training data emails, only 100 are labelled as spam. All other training data are instances of non-spam. What is the problem in training your model with such an unbalanced training data set? How will you avoid this issue?
11. In natural language processing, neural sequence to sequence models can be either recurrent neural network models such as LSTMs/GRUs or they can be transformer models. What are the advantages and disadvantages of each of these two model families?
12. You are trying to build a text classification model. Your friend suggests that for encoding the text, instead of using an RNN model, you should use a convolutional neural network (CNN) model. Do CNN models work well on text? What are the advantages and disadvantages?
13. Can you explain the concepts of self-attention and cross-attention?
14. In training deep learning neural networks, what is catastrophic forgetting? How does one address this issue?
15. Can you explain the difference between model parallelism, data parallelism and tensor parallelism? Give examples of recent training models that employ one or more of these approaches?

Feel free to reach out to me over LinkedIn/email if you need any help in your coding interview preparation. If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Wishing all our readers happy coding until next month! Stay healthy and stay safe. END

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